Intelligent Driver Alert System on Pothole Detection using Deep Learning Techniques

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Abstract

The detection of potholes is an important issue that proves to be a risk to drivers, so identifying and alerting them in a timely manner is crucial. Potholes can cause severe damage to vehicles and thus in-order to prevent them, we have proposed a method demonstrating a solution for detecting potholes using deep learning techniques to alert the driver after detection of the potholes. Our project aims to implement the object detection technique integrated with Convolution Neural Network technique and the You Only Look Once algorithm to efficiently train our model with a large dataset where the model is able to detect and classify whether the road is filled with potholes and provide a warning to the driver. We use Open Source Computer Vision Library for live detection used for extracting features and images from videos and storing them. The deliverable will hence be an accurate model that makes a prediction of potholes and the result will be the bounding boxes that appear around it that prove the detection. The aim of this paper is to provide an extensive overview of the challenges and opportunities in the field of pothole detection.

1. Introduction

Potholes are a common and persistent problem on roads, causing damage to vehicles and posing a safety hazard to drivers. The necessity for effective and efficient pothole detection technologies is becoming more and more critical due to the growing number of vehicles on the road and the deteriorating state of the road infrastructure. Traditional manual inspection methods are time-consuming and can lead to missed or poorly repaired potholes. A rising number of research projects are concentrating on the creation of novel pothole-detecting technologies, such as computer vision and machine learning, to address this problem. These techniques have a great deal of potential for increasing pothole identification speed and accuracy while also lightening the workload associated with manual inspection.

There are various approaches to pothole detection because it is considered an area of extensive research and development of new techniques. Some of the pre-existing techniques for detecting potholes are: vibration method, traditional method and 3D reconstruction based method.

The vibration method makes use of sensors known as acceleration sensors that detect the pothole based on the speed of the vehicle. Even-though this technique is proved to be cost effective, one of its major drawbacks is that it cannot detect the pothole in prior and alert the driver to avoid it. The detection occurs only when the vehicle reaches the pothole. Traditional methods on the other hand require a committee to investigate and analyse recordings of detected potholes which is inefficient and time consuming which is why we use the real time detection method using OpenCV to overcome this drawback.

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A method based on 3D reconstruction draws a prediction based on the shape of pits and measures their volume based on stereo vision technology. This is considered to be highly accurate but requires the use of costly equipment such as kinetic sensors and infrared technology only for short range detection.

With an emphasis on the benefits and drawbacks of various strategies, we seek to present an overview of the status of pothole-detecting techniques in this study. The objective of this study is to provide insights into the challenges and opportunities in the field of pothole detection and to inspire further research and innovation in this area. Our proposed system would be designed in an attempt to provide the driver with beep sounds or notifications to alert him and guide him in the right direction to avoid potholes and vehicle damage.

2. Materials and Methods

The proposed system works with the help of a very simple and user-friendly GUI which provides you with two options, “Start” and “Stop”, and helps you achieve real-time potholes detection using the YOLO algorithm. When one selects the “Start” Option, the Camera switches on and live detection of potholes takes place. Now the camera keeps recording while the YOLO algorithm works in the back end simultaneously. The required information necessary for detecting the potholes is captured. The images are extracted throughout the live recording and accordingly, a detection process is carried out.

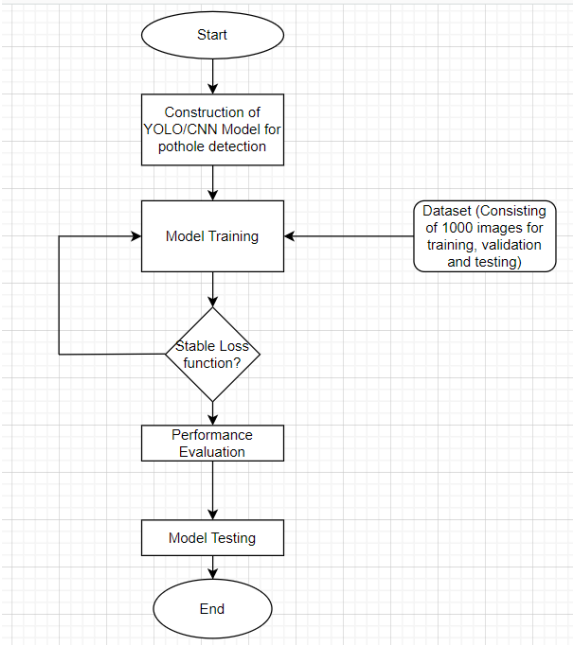

FIGURE 1. **Flowchart**

The YOLO algorithm takes these images one by one and processes them. It compares the information recorded from the images with the YOLO weights. When a pothole is detected, it predicts a bounding box around it. The bounding box consists of a box label and Predicted Accuracy Percentage where Accuracy Percentage refers to the percentage of how accurately a pothole is detected. Whatever number of potholes are detected in each image gets reflected on the console output. As you select the “Stop” option, the system exits the GUI and thus the detection process stop

Model Architecture

Pothole detection is a technology-driven solution for identifying and locating potholes on roads and highways. The objective of this project is to accurately recognize potholes in images or video frames with minimal false positive or false negative results. By achieving these objectives, pothole detection has the potential to improve the overall quality of roads and to provide a safer and more efficient approach to road maintenance, also can help prioritise repair efforts and improve road safety by reducing the risk of vehicle damage and accidents.

The project will involve the use of computer vision and deep learning techniques to develop a system that can detect potholes in real-time. The system will process images or video frames from cameras and use algorithms such as Convolutional Neural Networks (CNNs) and You Only Look Once (YOLO) to detect and locate potholes. The system will be trained on a large dataset of images of roads with and without potholes to improve its accuracy and performance.

The last step to the detection process would include the generation of an alert sound which we plan on implementing using either of the two ways. Using the Pygame module to produce sound in Python with the use of audio files which would be present on our system or by directly using a program file to run.

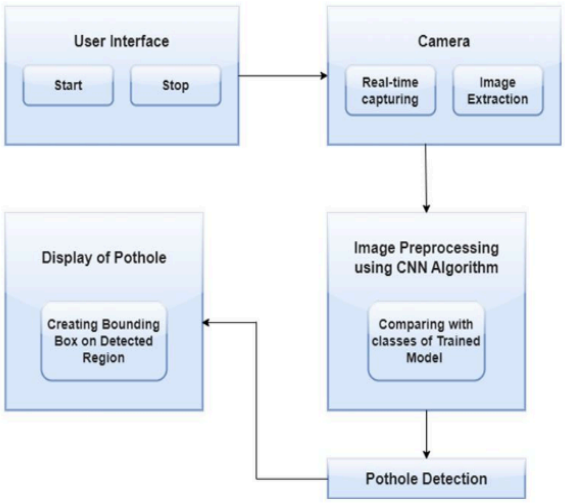

FIGURE 2. **Model Architecture**

In Android Studio, you can create a ToneGenerator class that generates simple sine wave tones using the Android built-in AudioTrack class. The class takes in the frequency of the tone to be generated during initialization. The class also creates an AudioTrack object with the specified sample rate and buffer size. You can customise the tone type, duration, and delay time to suit your needs. This can be useful for producing beep sounds or alert messages in your Android application.

The end result of this project will be a practical and effective solution for pothole detection that can be used by road maintenance agencies and organisations to improve the safety and quality of roads and highways.

3. Results and Discussion

A) Importance

Despite efforts to maintain roads and prevent the formation of potholes, they remain a persistent problem in many regions, causing damage to vehicles and posing a safety hazard to drivers. Traditional manual inspection methods are time-consuming and can result in missed or poorly repaired potholes. There is a need for more efficient and effective methods for pothole detection that can improve road safety and maintenance operations.

The challenge is to develop and implement new technologies for pothole detection that are accurate, efficient, and scalable. This involves exploring the use of computer vision, machine learning and deep learning to automate the detection process, as well as evaluating the impact of these methods on road maintenance operations and the well-being of road users.

B) Interpretations of the results

A large dataset of images was collected which was used to train and test the model. The degree of accuracy indicates how well a model is performed and its ability to accurately detect the potholes successfully while the loss parameter is the performance depreciation during the training phase. For a model to detect objects successfully the accuracy parameter should be high and the loss parameter should be the lowest.

However, thermal imaging, which has various advantages over other approaches like greater accuracy, cheap cost, and less environmental impact, has never been employed for pothole identification before. The complex does not require the risk of driving over potholes and can operate at night or in foggy weather.

C) Literature Review

The literature studies that were referred to, indicated a variety of deep learning techniques used for the detection of potholes. These techniques prove to produce accurate results when compared to current prediction models.

In this field, a lot of work has been done worldwide. The outcomes appeared to be beneficial. The authors: S. Ezekiel1, O. E. Taylor2* and D. J. S. Sako3 in [1] presents a smart system for pothole detection using computer vision with transfer learning. A trained model is used to categorise the actual occurrences of potholes using data from smartphone sensors, specifically the accelerometer, and gyroscope.

However in [2] the authors: Shitij Pawar, Siddhi Jagtap proposed an architecture consisting of four primary phases namely data acquisition, data pre-processing, model building, and classifier evaluation. The raw data was collected using the smartphone's built-in accelerometer and gyroscope sensors to build the datasets. It has been proposed that a neural network model was trained and rigorously tested in order to classify potholes precisely with accuracy by reducing the complexity as well as the time taken.

Authors: Shubra Rao Kuthyar, Sahana Manjesh and Rahul Davis proposed in [3] a Pothole Detection and an Alerting System using Mobile Sensors and Deep Learning in which the key findings, mentions that the architecture of the system consists of 4 primary stakeholders namely the Public and people, The Elected Representative (ER), the Public Works Department (PWD), and the Contractor. Potholes are reported by the general public using the public mobile application.

In [5] author : Khaled R. Ahmed proposed an approach in the paper involving developing CNNs with deep learning techniques to detect potholes in real-time with sufficient accuracy, achieved by introducing a modified VGG16 (MVGG16) network that reduces computational costs and improves training results through the removal of convolution layers and use of different dilation rates.

D) Result conclusion

Our study and implementation followed a systematic approach where during the training phase, the number of epochs was increased in order for the accuracy parameter to increase and the loss parameter to decrease to improve the efficiency and performance of our model. This is done using the call back function that retrieves the parameter values like the accuracy of the loss for each epoch for plotting a graph and analysing it. A call back function is an object that can operate various stages of the training.

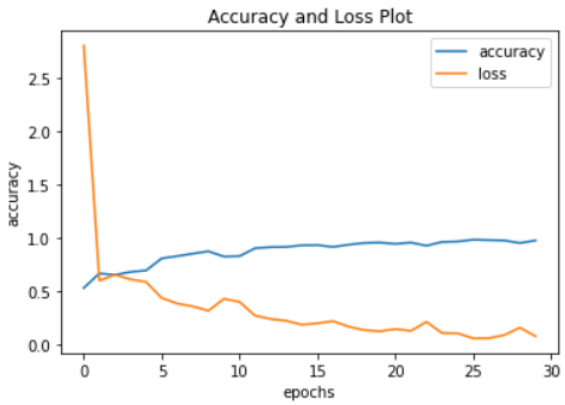

FIGURE 3. Training Accuracy and loss Graph

FIGURE 4. Testing Accuracy and loss Graph

Conclusion

Our project aims to efficiently train our model on a large dataset such that the model is capable of differentiating between smooth roads and the roads that have potholes in-order to provide a warning to the driver indicating that there is a pothole. In the

future, we propose to develop an application known as 'Intelligent Driver Alert System on Potholes' which assists the driver in avoiding potholes on the roads, by giving prior warnings by using mobile sensors to alert the driver.

Artificial intelligence techniques for pothole detection can aid in better road repair, particularly in resource-constrained poor nations. Our proposed pothole detecting system's accuracy was 91.27% utilising a CNN-based model.

Moreover, this research can be expanded to identify the location and communicate to the government to fix the potholes which would be our future scope.

FIGURE 5.. Detected pothole from YOLOv3 model

Authors' Note

All our authors declare that there is no conflict of interest regarding the publication of this article. The aim was to keep the paper free of plagiarism.

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